

Algorithmic Trading Enhanced by AI: Using News, Geopolitical Events and NLP-Based Sentiment in Forex Decision-Making

The use of algorithmic trading has expanded significantly over the past decade and a half, driven by advances in computational power, data availability, and artificial intelligence. My first experience with automated trading dates back to around 2010, when I collaborated on the development of a Forex day-trading system built on technical analysis and integrated directly with MetaTrader. Although the system initially produced encouraging results, its performance deteriorated over time, highlighting a persistent problem in models that rely exclusively on graphical patterns: they fail to capture the broader informational forces that drive modern financial markets. This limitation remains central to the theme of this work, which investigates how algorithmic trading can be enhanced by integrating external sources of information—particularly geopolitical events, global conflicts, macroeconomic announcements and sentiment extracted from text.

Since 2010, global markets have become increasingly sensitive to events that lie far beyond historical price data. Wars, diplomatic crises, sanctions, supply-chain disruptions and disputes involving critical minerals have reshaped economic relations and generated abrupt market swings. Decisions taken by heads of state and unexpected policy changes also influence investor expectations in ways that cannot be captured by chart patterns alone. Episodes such as the impactful reactions to political shifts in the United States illustrate how markets respond more intensely to geopolitical narratives than to technical indicators. This evolution in market behavior, combined with the exponential growth of digital content, motivates a renewed attempt to develop a trading system capable of interpreting these complex signals.

In parallel, technological progress has been remarkable. Techniques in Natural Language Processing (NLP) and sentiment analysis now allow the automated extraction of insights from news articles, institutional reports and social media discussions. More recently, Large Language Models (LLMs) have introduced new capabilities for contextual understanding and event interpretation, enabling automated systems to identify not only what happened, but also the likely implications of political and economic developments. The availability of these tools, combined with the limitations experienced in my earlier project, forms a strong justification for revisiting the Forex domain with a modern approach that reflects today's informational richness and technological sophistication.

Based on this context, the general objective of this work is to develop and evaluate a hybrid algorithmic trading strategy that integrates traditional technical indicators with sentiment-based signals derived from textual analysis, including geopolitical information and macroeconomic events. By combining numerical market data with insights extracted from contemporary AI techniques, the intention is to create a more adaptive, robust and context-aware framework capable of operating in a financial environment shaped by global instability and rapid information flow. This work therefore represents both a continuation of a project initiated in 2010 and an effort to align algorithmic trading with the demands and opportunities of the current AI-driven world.